Precision Emotion Regulation for Mood Disorders: A Micro-Expression-Based Closed-Loop Music Therapy System

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Abstract

In recent years, individuals have been experiencing increasing levels of emotional stress, and emotion-related disorders have gradually become a major factor affecting mental health. Music, due to its natural capacity for emotional regulation, has demonstrated unique advantages in assisting psychological intervention. However, most existing music therapy systems remain at the stage of static recommendation and lack the ability to perceive users' real-time emotional states and respond dynamically.To address this limitation, we design and implement a Micro-Expression-Based Closed-Loop Music Therapy System. This system consists of two key modules: (1) a micro-expression emotion recognition module that utilizes deep neural networks to perform real-time emotion classification based on users' facial micro-expressions during music listening; and (2) a music generation module, which incorporates an alignment network (Align-Net) to align the micro-expression image embeddings with the CLAP-encoded personalized music descriptions, and generates corresponding music using a Latent Diffusion Model (LDM).To evaluate the feasibility and effectiveness of the proposed system, we conducted user studies assessing both emotion recognition accuracy and the quality of generated music. Experimental results demonstrate that the system enables emotion-driven dynamic music generation and provides a practical foundation for building more intelligent and responsive music therapy technologies.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**.

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Music Therapy,Music Generation,Emotion Recognition, Micro-expression,Deep Learning

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1 Introduction

In modern society, with the acceleration of life's pace and the intensification of social competition, people are experiencing significantly increased psychological stress [29]. The incidence of anxiety, depression, and related mental health issues continues to rise [33]. This trend not only affects individuals' quality of life but also imposes a heavy burden on public healthcare systems.

To alleviate psychological stress, the current mainstream treatments include pharmacological therapies and non-pharmacological interventions. Although pharmacological therapies can relieve symptoms to a certain extent, they are often accompanied by problems such as dependency and side effects [15]. Non-pharmacological interventions, such as music therapy, have been widely studied and applied; their natural and non-invasive characteristics have been proven effective in regulating emotions and promoting mental health [8, 11, 22–24, 28]. Studies have shown that applying music therapy specifically tailored to users and aligned with their current emotional states can significantly enhance emotional well-being [3, 11]. To achieve the best therapeutic outcomes, it is crucial to use music that is both personalized and emotion-adaptive [1, 2]. In today's increasingly stressful society, people of different personalities, occupations, and nationalities are all facing mental health challenges [29, 33]. Large-scale promotion of emotion-regulating approaches such as music therapy is considered an effective means to reduce the incidence of mental disorders and alleviate societal stress [5, 10, 26].**However, existing music therapy systems still have many limitations and shortcomings.** First, they often require the involvement of professional music therapists, making large-scale and low-cost

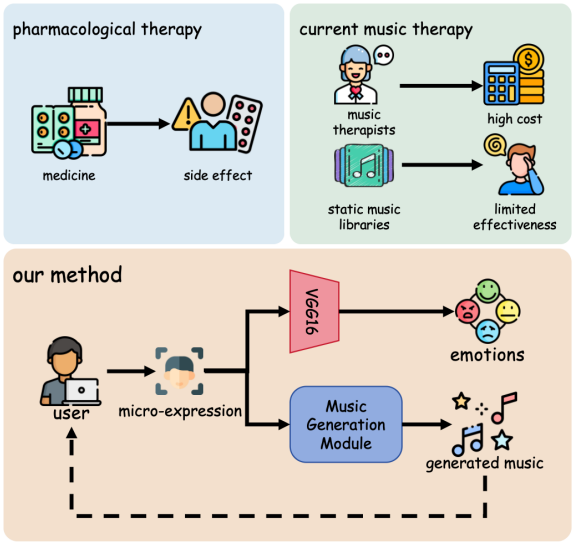


Figure 1: This figure illustrates the shortcomings of existing emotional healing methods and the general framework of our approach. Pharmacological therapy may bring side effects to patients, while current music therapy is often costly and unable to generate personalized music based on users' emotions. Our method can dynamically detect user emotions and generate more fine-grained music to achieve better emotional improvement effects.

applications difficult [12].In developmental programs for hospitalized children [16], hospitals employ highly skilled and rigorously trained music therapists to deliver music therapy. However, for most users, accessing professional music therapists affordably and on demand remains extremely difficult.Second, most current systems lack the capability to dynamically detect users' emotions and adapt the music in response to an individual's real-time state, which makes it impossible to promptly identify the user's true emotional condition and generate corresponding personalized music [30]. Instead, they rely on pre-defined music libraries [6], which further limits their therapeutic effectiveness. Additionally, the use of static music libraries for therapeutic purposes may carry significant copyright infringement risks [19].Therefore, there is a pressing need for more accessible and personalized music therapy systems capable of detecting users' emotional states and generating individualized music.

Benefiting from the ability of micro-expressions to authentically reflect human emotions, we propose a Micro-Expression-Based Closed-Loop Music Therapy System. The incorporation of micro-expressions enables our system to automatically and continuously monitor users' genuine emotional states in real time, providing a foundation for user status assessment and subsequent adjustment of therapeutic plans.This system consists of two main components: a micro-expression emotion recognition module, which recognizes human emotions in real time from captured facial images, and a music generation module, which creates therapeutic music conditioned on the detected micro-expressions.We aim to realize a more fine-grained and personalized music therapy process without the involvement of professional music therapists. To demonstrate that our music therapy system is accessible and effective, we conducted a user study involving 10 participants. It is our expectation that this system can provide inspiration and guidance for the future development of psychological health and music therapy systems. Our main contributions can be summarized as follows: Our main contributions can be summarized as follows:

(1) We propose a novel micro-expression-based closed-loop music therapy system that recognizes users' emotions through micro-expression analysis and dynamically generates music accordingly, achieving a truly adaptive and personalized therapeutic process.
(2) We designed a music generation module based on micro-expressions. By leveraging large language models with visual understanding capabilities, we generated diverse training samples to assist the model in generating fine-grained music.
(3) We designed and conducted a user study to provide evidence demonstrating the feasibility and effectiveness of the music therapy system. As a result, we also collected a more diverse set of micro-expression samples and user feedback, which can help improve the system in future work.

2 Related Works

2.1 Music Therapy

Music therapy, as a non-pharmacological intervention, has shown significant therapeutic effects in various fields such as mental health and neurorehabilitation in recent years. It plays a positive role in emotional regulation, anxiety relief, depression management, stress reduction, and physiological stabilization. The American Music Therapy Association defines music therapy as a structured, evidence-based clinical practice that promotes cognitive, emotional, and physical well-being through music-based activities [8].

In terms of anxiety reduction, Lu et al. conducted a meta-analysis of randomized controlled trials and found that music therapy significantly reduced anxiety levels across diverse populations [22]. Nilsson's systematic review further supported these findings, demonstrating that music interventions effectively alleviate both anxiety and pain in perioperative and clinical care settings [28]. During the COVID-19 pandemic, Giordano et al. showed that even a single session of receptive music therapy could markedly reduce anxiety and improve vital signs among hospitalized patients [11].

Music therapy also shows promise in treating depression. Aalbers et al. demonstrated its general effectiveness in alleviating depressive symptoms [1], and in a subsequent study, developed an improvisational music therapy protocol tailored for young adults with depression, highlighting its flexibility and clinical value [2].

For individuals with post-traumatic stress disorder (PTSD), Carr et al. conducted a randomized controlled trial and found that group music therapy helped reduce trauma-related symptoms, and improved self-expression and social functioning [3]. In the field of neurorehabilitation, Magee et al. emphasized the importance of integrating music therapy into rehabilitation teams, particularly for patients with acquired brain injuries, showing its effectiveness in supporting motor, language, and emotional recovery [23, 24].

However, music therapy still has numerous shortcomings and limitations. For instance, music therapy systems involving experts often require substantial financial expenditures; individuals exhibit significant variations in their preferences for musical rhythm and style, and static music therapy systems may compromise intervention efficacy, potentially eliciting resistance in some patients. Furthermore, existing music therapy systems lack sufficient dynamic attention to users' emotional states, making it difficult to recognize their emotions and adjust musical styles accordingly. Monotonous music selection may lead to user fatigue or discomfort.

2.2 Emotion Recognition With Micro-Expression

Emotion is defined as a neurophysiological and cognitive change triggered by internal or external stimuli [31]. Both the physical health and mental well-being of humans can be influenced by emotions and are reflected through emotional states. The core objective of emotion recognition lies in detecting human emotional states [14].

According to [37], micro-expressions are defined as brief facial emotional displays typically lasting between 40-500 milliseconds that unconsciously leak when individuals attempt to suppress genuine emotions. Micro-expressions have three key characteristics: (1) the transience and spontaneity of micro-expressions make them remarkably difficult to fake; (2) significant individual differences exist in micro-expression patterns under identical emotional states; and (3) expression duration shows an inverse relationship with emotional intensity. These properties establish micro-expressions as highly reliable indicators in emotion recognition, particularly valuable for real-time detection of users' genuine emotional states in human-computer interaction music therapy.

Due to the outstanding characteristics of micro-expressions (MEs), a wide range of machine learning and deep learning methods have been applied to micro-expression recognition (MER) tasks. Zeng et al. [34] conducted a comprehensive survey of recent methods over the past five years, including datasets, preprocessing techniques, evaluation metrics, and recognition accuracy, and analyzed strategies based on LBP, optical flow, and deep learning approaches.Zhang et al. [38] proposed *MicroEmo*, a novel framework based on multimodal large language models (MLLMs), which focuses attention on localized facial micro-expression dynamics and the context dependency of utterance-aware video clips. MELLM was also introduced, which leverages the reasoning ability of MLLMs combined with fine-grained facial motion perception [39].Given the powerful cross-modal semantic alignment capability of the CLIP model, Liu et al. [21] incorporated this into MER by proposing a new method called *MER-CLIP*, which enhances high-level understanding and captures subtle relationships between MEs and emotions.To address the challenges of limited MER data and significant individual differences, Wang et al. [34] proposed a dual-branch meta-auxiliary learning method named *LightmanNet* for fast and robust micro-expression recognition.Zeng et al. further introduced *HTNet*, a Transformer-based framework that effectively identifies key regions of facial muscle movement, achieving promising results on MER tasks. Meanwhile, Nguyen et al. [27] presented *Micron-BERT*, a novel approach to facial micro-expression recognition, which automatically captures such movements in an unsupervised manner based on two key mechanisms.The aforementioned studies exhibit competent capabilities in MER tasks, which substantiates the feasibility and reliability of inferring authentic emotions through micro-expressions.

2.3 Music Generation

With growing demands from music enthusiasts, artists, and music therapy applications, an increasing number of music generation models have emerged in the community.Based on the outstanding performance of transformer, Copet et al. introduced MUSICGEN, comprised of a single-stage transformer LM together with efficient token interleaving patterns, which eliminates the need for cascading several models, e.g., hierarchically or up sampling [7]. Furthermore, diffusion models have achieved remarkable results in image generation tasks, which has aroused more scholars' interest in whether diffusion models can be applied to music generation tasks. Based on AudioLDM [20],Chen et al. first constructed a state-of-the art text-to-music model, MusicLDM, that adapts Stable Diffusion and AudioLDM architectures to the music domain, in order to overcome the challenges due to limited availability of music data and sensitive issues related to copyright and plagiarism [4]. Melechovsky et al. proposed Mustango: a music-domain-knowledgeinspired text-to-music system based on diffusion, which includes the LDM , a MuNET consists of a UNet and multiple conditioning cross-attention blocks, and encoders [25]. These music generation models have achieved excellent results in text-to-music generation tasks. However, it is insufficient that few models currently focus on how to generate personalized music based on human emotions—such as the emotions recognized by ME—and apply them to music therapy systems.

3 Methodology

In this section, we will introduce the structure of our closed-loop music therapy system, including two main parts, the Micro-Expression Emotion Recognition Module and the Music Generation Module. The former takes VGG16 as the backbone, aiming to recognize human true emotions through micro-expressions [32]. The latter consists of a text-image alignment module(Align-Net) and a music generator with LDM. In the subsequent content of this section, we will introduce the specific architecture and functions of these modules in a sequential manner.

3.1 Micro-Expression Emotion Recognition

Inspired by [32], we leverage the VGG16 to classify emotions by micro-expressions. The structure and parameters of VGG16 is illustrated in Table 1.

VGG16 is a deep convolutional neural network designed with the principle of using multiple small-sized convolutional filters to construct a deep architecture. This design enhances the model's representational capacity while keeping the number of parameters manageable. The network takes a $224 \times 224 \times 3$ RGB image as input and processes it through five sequential convolutional blocks. Each block consists of several convolutional layers with 3×3 filters activated by the ReLU function. To maintain the spatial resolution of the feature maps, these filters are applied with a stride

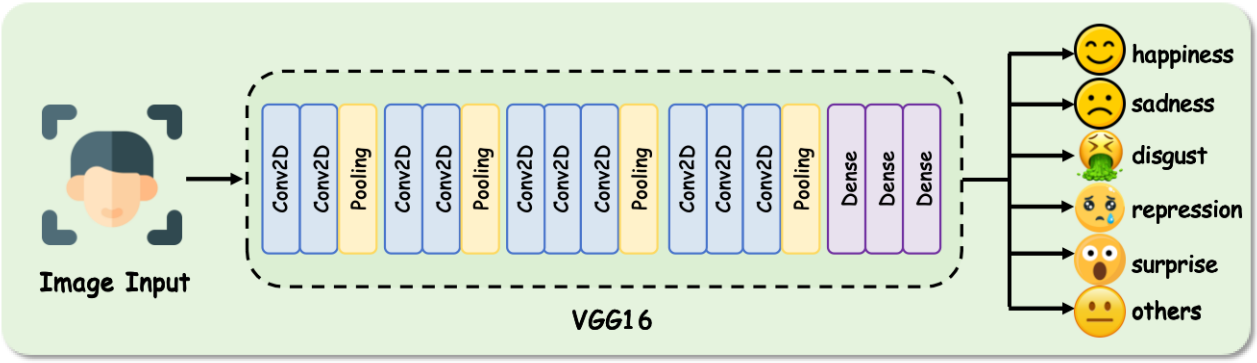


Figure 2: The architecture of our micro-expression emotion recognition model is as shown. We employ VGG-16 to perform emotion recognition from micro-expression images. .

Table 1: Architecture of VGG16, which includes 13 convolutional layers and 3 fully connected layers. The model achieves strong image recognition performance with only 134.29 million parameters.

Layer (type)	Output Shape	Param #
input_layer	(Batch Size, 224, 224, 3)	0
block1_conv1	(Batch Size, 224, 224, 64)	1,792
block1_conv2	(Batch Size, 224, 224, 64)	36,928
block1_pool	(Batch Size, 112, 112, 64)	0
block2_conv1	(Batch Size, 112, 112, 128)	73,856
block2_conv2	(Batch Size, 112, 112, 128)	147,584
block2_pool	(Batch Size, 56, 56, 128)	0
block3_conv1	(Batch Size, 56, 56, 256)	295,168
block3_conv2	(Batch Size, 56, 56, 256)	590,080
block3_conv3	(Batch Size, 56, 56, 256)	590,080
block3_pool	(Batch Size, 28, 28, 256)	0
block4_conv1	(Batch Size, 28, 28, 512)	1,180,160
block4_conv2	(Batch Size, 28, 28, 512)	2,359,808
block4_conv3	(Batch Size, 28, 28, 512)	2,359,808
block4_pool	(Batch Size, 14, 14, 512)	0
block5_conv1	(Batch Size, 14, 14, 512)	2,359,808
block5_conv2	(Batch Size, 14, 14, 512)	2,359,808
block5_conv3	(Batch Size, 14, 14, 512)	2,359,808
block5_pool	(Batch Size, 7, 7, 512)	0
flatten	(Batch Size, 25088)	0
fc1 (Dense)	(Batch Size, 4096)	102,764,544
fc2 (Dense)	(Batch Size, 4096)	16,781,312
predictions (Dense)	(Batch Size, classes)	24,582

of 1 and padding of 1. Each convolutional block is followed by a 2×2 max-pooling layer, which reduces the spatial dimensions and introduces translation invariance. After the convolutional and pooling stages, the feature maps are flattened and passed through three fully connected layers, with the final layer using a Softmax activation function to produce class probabilities.The complete architecture of this module is illustrated in Figure 2.

3.2 Alignment Between Text and Image

To fully leverage the emotional information conveyed by human micro-expressions, we directly generate image embeddings from micro-expression images and use them as conditional vectors for music generation. We propose to train an Align-Net to semantically align the micro-expressions with the target music description text (the generation process of the target music descriptions will be discussed in detail in section 4).

Specifically, we encode the music descriptions using CLAP (Contrastive Language-Audio Pretraining) to obtain the textual embedding $E_t \in \mathbb{R}^{1\times T}$, where T denotes the dimension to which the final text embedding is projected. Due to CLAP's strong cross-modal capabilities, we assume that E_t captures the semantic content of the textual descriptions. Music generated based on E_t is thus expected to have a positive effect on emotional regulation. For micro-expression image processing, we take a $24 \times 24 \times 3$ RGB image I as input and use `ResNet50` for initial feature extraction [13]. The core of ResNet lies in the introduction of residual learning, which addresses the degradation problem caused by vanishing/exploding gradients in traditional deep neural networks during training. By incorporating skip connections, the model no longer directly fits the complex nonlinear transformation $H(x)$, but instead learns the residual mapping between input and output:

$$F(x) = H(x) - x \tag{1}$$

In our adopted ResNet50 architecture, the bottleneck structure is employed to reduce computational complexity, enabling the model to capture spatial features while maintaining relatively low computational overhead.We emphasize that in our architecture, the role of ResNet50 is to extract features from micro-expression images, rather than its original purpose of performing classification tasks. Therefore, we replace the final fully-connected layer of ResNet50 with an identity layer, using the extracted image features $\mathbf{F}_i \in \mathbb{R}^{1\times M}$ as the final output, where M is set to 2048 in ResNet.To meet the requirements of the alignment task, we employ a projection layer to map the image features into the same dimensional space as the text embeddings obtained from CLAP encoding, yielding the final image embeddings $\mathbf{E}_i \in \mathbb{R}^{1\times T}$.And the complete alignment pipeline

can be formulated as follows:

$$\mathbf{E}_i = \text{Proj}(\text{Res}(\mathbf{I})) \tag{2}$$

where $\mathbf{E}_i$ denotes our image embedding, Proj represents the projection layer that maps vectors from $\mathbb{R}^{1\times M}$ to $\mathbb{R}^{1\times T}$, Res denotes our modified ResNet model (with image features as final output), and $\mathbf{I}$ is the input micro-expression image.We employ the mean squared error (MSE) as our loss function to train the modified ResNet50 and projection layer defined above. The optimization process continuously aligns $\mathbf{E}_i$ with $\mathbf{E}_t$, thereby enabling the image embeddings generated from micro-expression images to capture the semantic meaning of music descriptions.

3.3 Music Generation With Micro-Expression Embedding

Due to the outstanding performance of MusicLDM [4] in text-to-music generation tasks, we select it as the backbone for our music generation module. MusicLDM has similar architecture as AudioLDM [20]:a contrastive language-audio pretraining (CLAP) module [35], a latent diffusion module with a pretrained audio variational auto-encoder (VAE) [17], and a Hifi-GAN neural vocoder [18]. CLAP is a contrastive language-audio pretraining model that aligns textual descriptions with audio content in a shared latent space through contrastive learning, enabling cross-modal semantic understanding.Pretrained Audio VAE compresses mel-spectrograms into low-dimensional latent variables during encoding, and reconstructs high-quality spectral representations during decoding. And the HiFi-GAN vocoder converts mel-spectrograms generated by the diffusion model into time-domain waveforms, achieving high-fidelity audio synthesis with enhanced perceptual quality.

In Latent Diffusion Models (LDM), there are two main processes. In the forward process, the original audio S is encoded by the Audio VAE into a more compact latent representation $z_0 = \text{Encoder}(S)$. To learn the generative process in this latent space, LDM applies a forward diffusion process to z_0, where Gaussian noise is gradually added, pushing its distribution closer to a standard normal distribution. This forward process is defined by the following Markov chain:

$$q(z_t \mid z_{t-1}) = \mathcal{N}(z_t; \sqrt{1-\beta_t}\, z_{t-1}, \beta_t \mathbf{I}) \tag{3}$$

where $\beta_t \in (0,1)$ is a predefined noise schedule parameter. It is worth noting that the objective of the forward diffusion process is to enable the model to learn how to progressively denoise z_t to recover the original latent representation z_0. However, in our model, the entire LDM is frozen, and thus the forward process is not actually performed.

In the denoising process, the model learns to progressively recover the latent vector z_0 from standard Gaussian noise z_N conditioned on information E. Specifically, in our experiments, the embedding vector E_i generated from the micro-expression image is used as the conditioning vector for the LDM:

$$p_\theta(z_{n-1} \mid z_n, E_i) = \mathcal{N}\left(z_{n-1}; \mu_\theta(z_n, n, E_i), \sigma_n^2 \mathbf{I}\right) \tag{4}$$

where the mean function μ_θ is predicted by the trained diffusion model based on the current latent variable z_t, time step t, and condition E_y. The variance σ_t^2 controls the randomness during sampling. The computation can be expressed by the following formulas:

$$\mu_\theta(z_n, n, E_i) = \frac{1}{\sqrt{\alpha_n}}\left(z_n - \frac{\beta_n}{\sqrt{1-\bar{\alpha}_n}}\epsilon_\theta(z_n, n, E_i)\right) \tag{5}$$

$$\sigma_n^2 = \frac{1-\bar{\alpha}_{n-1}}{1-\bar{\alpha}_n}\beta_n \tag{6}$$

After the LDM generates the final latent representation z_0 through reverse diffusion conditioned on E_i, z_0 is decoded into a mel-spectrogram by the pretrained Audio VAE decoder, and then converted into the final target audio by the HiFi-GAN vocoder.The alignment and music generation process is illustrated in Figure 3.

4 Experiment

4.1 Experimental Setup

In this subsection, we introduce the training configurations of the micro-expression emotion recognition module and the align-net, as well as the method for generating target music descriptions.

First, for the training of the Micro-Expression Emotion Recognition module, we use the CASME II dataset [36]. The data used for training contains six emotion categories: disgust, happiness, repression, sadness, surprise, and others. We split the dataset into training and test sets with a ratio of 4:1.We use the pretrained VGG16 weights as the initial parameters and perform two-stage training: (1) freezing the feature extraction network and fine-tuning only the classification layers; (2) unfreezing the backbone and training the entire network. More detailed parameter configurations are provided in Table 2.

Table 2: Training configurations for Micro-Expression Emotion Recognition module

Configuration	Frozen Training	Unfrozen Training
Epoch	3	2
Learning Rate	1×10^{-3}	1×10^{-4}
Optimizer	Adam	Adam
Loss	crossentropy	crossentropy
Decay Rate	0.94	0.94
Batch Size	32	16

As for the training of the align-net, as mentioned above, we aim to align the embedding vectors of micro-expression images with the CLAP-encoded representations of the target music descriptions. In this section, we focus on how to generate personalized music descriptions based on a large language model(LLM) capable of multimodal understanding.First, we provide the six aforementioned emotional states one by one as the user's current condition and prompt the LLM to generate a music description D_i for each state with no more than 20 words. Then, to enable more fine-grained music generation, we further generate a music description for each of the 8,748 micro-expression images in the training set. Specifically, for a given micro-expression image I and its corresponding emotion i, we instruct the LLM to modify the previously generated description D_i based on the user's current expression I, resulting in a more diverse and personalized description D_i'. Finally, we train the align-net to align the embedding vector of the micro-expression image I with the CLAP-encoded representation of its corresponding

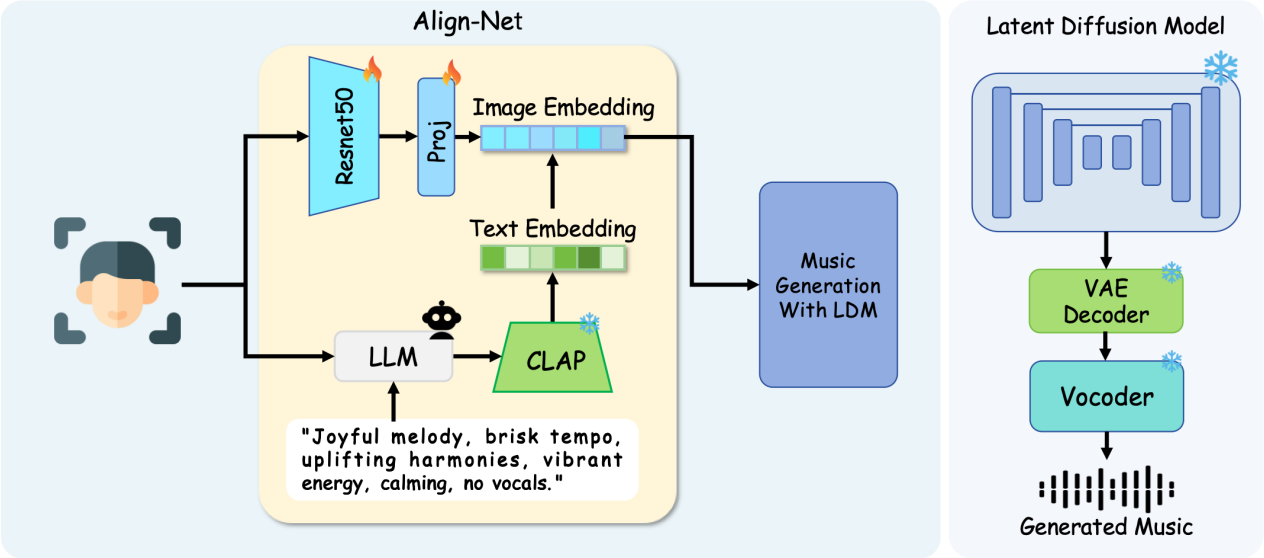


Figure 3: The architecture of our music generation module is as shown. This module consists of an Align-Net for aligning the micro-expression embedding vector with the CLAP-encoded target text, a Latent Diffusion Model (LDM), a VAE decoder, and a vocoder. The latent representation generated by the LDM, conditioned on the alignment vector, is subsequently processed by the VAE decoder and the vocoder to convert it into human-perceivable music.

Table 3: Training configurations for Align-Net

Configuration	Align-Net
Epoch	10
Learning Rate	3×10^{-4}
Optimizer	Adam
Loss	MSEloss
Batch Size	32

description D_i'.Similarly, the training configurations for align-net are provided in Table 3.

4.2 User Study

A user study was conducted to evaluate the feasibility and effectiveness of the proposed micro-expression-based closed-loop music therapy system.The overall procedure of our user study is illustrated in Figure 4.Our user study invited ten users and consists of five subsections. In the first subsection, participants are required to listen to real music while their micro-expression photos are captured during the listening process. These photos are then input into both the micro-expression emotion recognition module and the music generation module to detect the user's emotional state and generate corresponding music. In the next subsection, the generated music is played back to the participant, and the same process—photo capture, emotion recognition, and music generation—is repeated. At the end of each subsection, participants are required to report their actual emotional state while listening to the music. After completing the entire process, participants are also asked to evaluate the overall system by answering the following questions: (1) Did the system help improve your emotional state? (2) How would you rate the overall quality of the generated music? (3) Can you clearly distinguish between the generated music and the real music? To demonstrate the system's positive impact on human emotions, we also designed a controlled experiment. In this experiment, instead of using users' micro-expressions as conditional embeddings to generate music, we pre-generated a large collection of positive music to form a static music library with MusicLDM and randomly selected tracks from it for the trials. Each user is required to test both our system and the static controlled system following the aforementioned procedure, and choose the preferred system between the two after all tests are completed. After conducting the user study, we are able to collect a more diverse set of labeled micro-expression samples, which can be used to further improve our system based on user feedback and ratings.

4.3 Results

Here, we will discuss the results of the aforementioned experiments one by one. First, we tested the accuracy of our micro-expression emotion recognition module on the test set divided based on CASME II. It is worth noting that due to the small number of samples in the sadness category (close to 1/40 of the total sample size) in both the training and test sets, we did not include the results of this category in the figure. The final classification confusion matrix is shown in Figure 5. The overall accuracy of classification is 54.6%. We found that the emotion of repression is frequently misclassified as happiness with a high probability. This may be because humans tend to activate the zygomatic major muscle during repression, forming a "social fake smile" [9].

The results of our user study are shown in Figure 6. The left portion (subfigures (a)–(e)) demonstrates the outcomes using

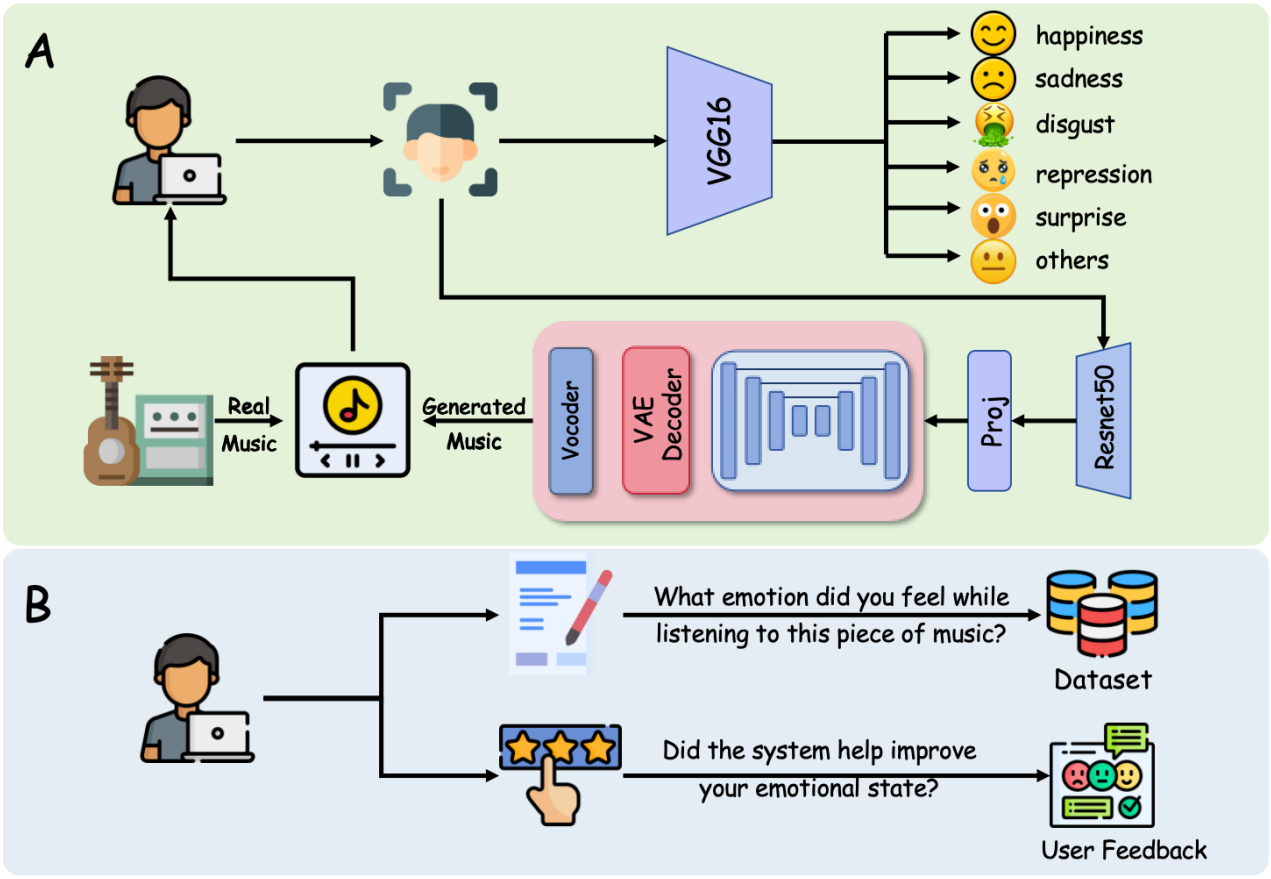


Figure 4: The overall framework of our User Study. Part A describes the specific process of the system recognizing emotions based on users' micro-expressions and generating music. Part B indicates the evaluation of the system by users after the completion of each subsection and the entire process.

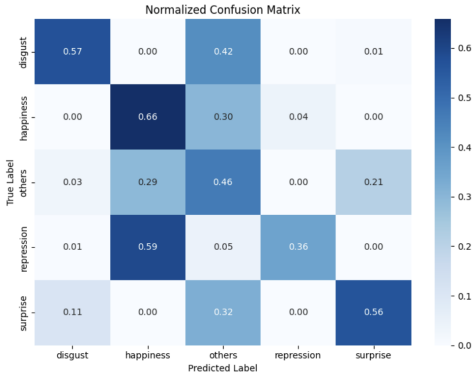


Figure 5: This figure shows the confusion matrix of the micro-expression emotion recognition module, with the sadness category excluded due to the extremely small sample size.The vertical axis represents the ground-truth emotions, while the horizontal axis indicates the emotions predicted by our model.

our proposed system, while the right portion (subfigures (f)–(j)) presents the therapeutic effects achieved with the pre-generated static music library. Specifically, subfigures (a) and (f) in Fig. 6 respectively demonstrate the authentic emotion distributions of all 10 participants under the two therapeutic systems. The five radar charts arranged left-to-right in each subfigure quantitatively measure emotional states during five consecutive therapeutic subsections. In the system we proposed (subfigure (a)), emotions transitioned from an initial random distribution (this was because users listened to randomly-emotion music in the first subsection) to a state where the majority of users were in a happy mood. This indicates that our system can guide people's emotions towards a positive direction to a certain extent. In the controlled experiment (subfigure (f)), there was indeed a tendency for users' emotions to improve initially. However, in the fourth and fifth subsections, the number of people in the "happiness" state decreased. This is consistent with our previously proposed conclusion that music that does not match human emotions cannot have a good therapeutic effect. Similarly, subfigures (b) and (g) show the distribution of user emotions predicted by our micro-expression emotion recognition model in the five subsections under the two systems. The average

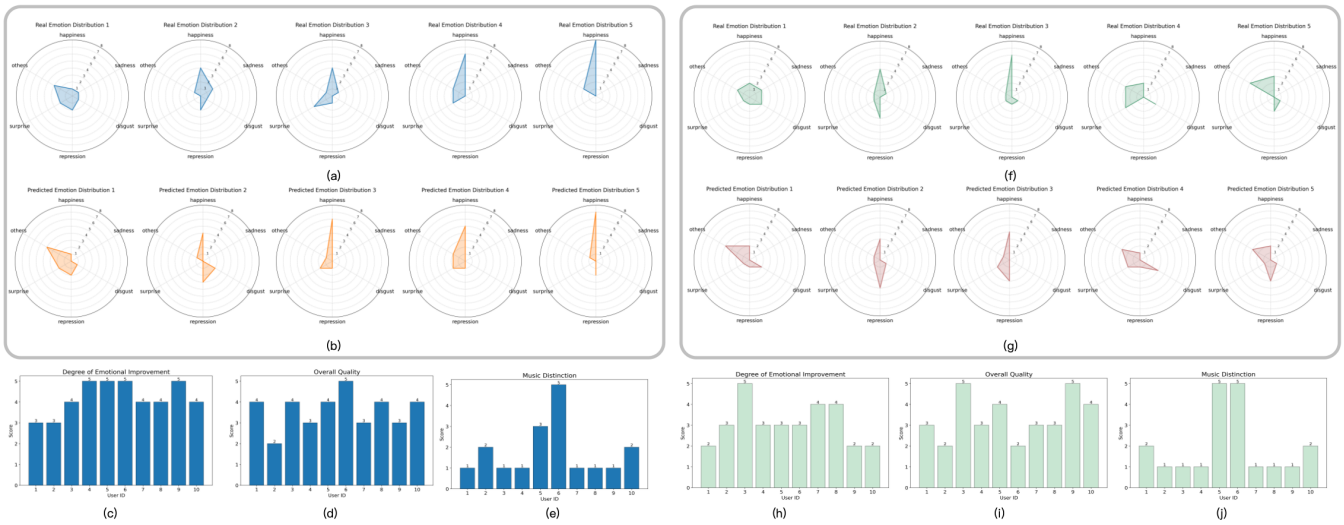


Figure 6: Results of our user study. The left half shows the experimental results using our proposed system, while the right half presents the results obtained with the pre-generated static music library.

prediction accuracy is 53.1%. After users completed the entire process of each system, we asked them to rate the system's degree of emotional improvement, overall music quality, and music distinction separately. Subfigures (c) and (h) show the degree of emotional improvement for each of the two systems (with 5 indicating the most significant emotional improvement). The former has an average score of 4.2, which is higher than the latter's average score of 3.1. This indicates that our method has a stronger ability to improve emotions compared to the static selection of music. Subfigures (c) and (h) show the overall generated music quality for each of the two systems (with 5 indicating the highest quality). Our system scored 3.6, slightly higher than the controlled group's 3.4. This may be because both systems generate music based on MusicLDM, resulting in similar overall quality but different styles and emotions. Subfigures (e) and (j) show the distinguishability of real music and generated music for the two systems (with 1 indicating difficulty in distinguishing), with average scores of 1.8 and 2.0, respectively. The similarity can also be explained by the aforementioned reasons. The preference distribution of users for the two systems is shown in Figure 7, with nine out of ten users considering our system to provide a better therapeutic effect.

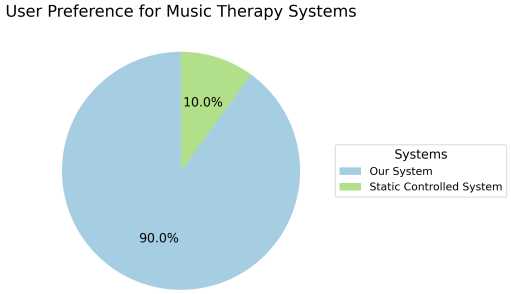


Figure 7: This figure shows the final preference distribution of users for the two systems.

5 Conclusions

In this paper, we propose a micro-expression-based closed-loop music therapy system to dynamically recognize users' emotions and generate therapeutic music based on micro-expressions. This system mainly consists of two parts: the micro-expression emotion recognition module and the music generation module, where the latter uses Align-Net to encode micro-expressions into embedding vectors as conditions for music generation. To evaluate the emotion recognition accuracy and emotion regulation effectiveness of the system, we conducted a user study and collected a more diverse set of micro-expression samples and user feedback. In future work, we aim to address several limitations, such as improving the generalization and accuracy of the emotion recognition module by leveraging newly collected micro-expression samples, and generating more diverse and higher-quality music descriptions through refined prompt engineering.In addition, regarding user study, we plan to expand the participant pool to demonstrate the system's generalizability, while incorporating more physiological indicators to reflect users' authentic experiences.

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